

JATIN LALCHANDANI

Robotics & Automation Engineering Student | AI/ML | Electronics & Robotics

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PROFESSIONAL SUMMARY

Robotics & Automation Engineering student with hands-on experience in machine learning, embedded systems, and PCB design. Skilled in building end-to-end projects spanning data-driven AI applications and hardware-based robotic systems. Actively developing technical expertise through independent projects, a student-led robotics initiative, and continuous exploration of autonomous systems.

EDUCATION

K. K. Wagh Institute of Engineering Education & Research

2025 – 2029

Bachelor of Technology (B.Tech.) in Robotics & Automation Engineering
Current GPA: 8.08 / 10

TECHNICAL SKILLS

Programming & Software	Python, Object-Oriented Programming (OOP), MATLAB
AI / Machine Learning & Data	Machine Learning, NumPy, Pandas, Matplotlib, Scikit-learn
Web Development	HTML, CSS, JavaScript, Flask, REST APIs / API Integration
Robotics & Electronics	Robotics, Electronics, Embedded Systems, PCB Design, Drone Technology, Sensors & Actuators, Motor Control, PID Control
PCB / CAD / Design Tools	KiCad, EasyEDA, Autodesk Fusion 360
Development Tools	Git, GitHub, Jupyter Notebook, VS Code
Other	Technical Writing, Leadership

PROJECTS

Stock Price Prediction Web Application

stock-price-prediction-a7vc.onrender.com

Python, Pandas, NumPy, Matplotlib, Scikit-learn, Flask

- Developed an educational machine learning application that predicts stock closing prices using historical stock data.
- Performed data preprocessing and exploratory analysis with Pandas and NumPy, then trained and evaluated regression models using Scikit-learn.
- Built and deployed a Flask-based web interface to serve model predictions, enabling users to visualize price trends online.

Line Follower Robot

ESP32, QTR-8A Sensor Array, N20 Motors, PID Control, EasyEDA, 3D Printing

- Designed and fabricated a custom PCB in EasyEDA to integrate an ESP32 microcontroller, motor driver, and QTR-8A sensor array on a 3S LiPo-powered chassis.
- Implemented PID control logic for real-time line tracking and tuned control parameters to improve path-following stability.
- Assembled 3D-printed mechanical components and integrated sensors, motors, and power systems into a functioning autonomous robot.

Smoke Detection Drone

Drone Technology, Robotics, Sensor Integration

- Worked on a drone-based platform concept for early smoke detection, aimed at supporting real-world fire-hazard monitoring.
- Explored sensor integration and autonomous flight fundamentals to combine robotics with practical environmental safety applications.

House Price Prediction

Python, Pandas, NumPy, Matplotlib, Scikit-learn

- Built a machine learning model to predict house prices using data preprocessing, exploratory data analysis, and regression techniques.
- Evaluated model performance using standard data science libraries in Python.

Rakhi AI Design Platform — In Development

Generative AI, Web Development, Frontend/Backend Integration

- Developing an AI-powered web platform that allows users to generate and customize festive Rakhi designs using generative AI.
- Building frontend and backend components to integrate AI-based image generation into an interactive web application.

LEADERSHIP EXPERIENCE

Founder & Lead

Present

Mecha Dojo — Student-Led Robotics & Electronics Initiative

- Founded and lead a student-driven initiative focused on robotics, electronics, and PCB design projects.
- Coordinate technical project work and mentor team members on robotics, embedded systems, and automation concepts.
- Explore emerging applications of AI, robotics, and embedded systems to drive hands-on learning within the group.

ACHIEVEMENTS & ACTIVITIES

- Built and deployed multiple independent machine learning and robotics projects spanning software and hardware domains.
- Designed and fabricated a custom PCB for an autonomous line-following robot, integrating sensors, motor control, and PID tuning.